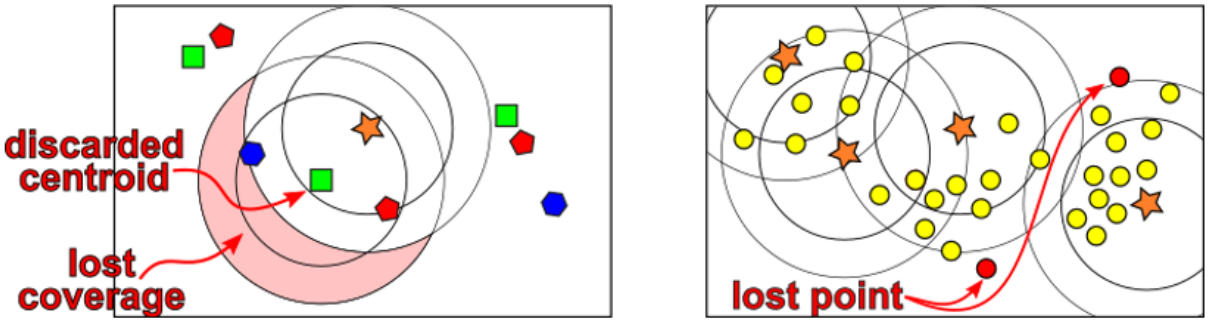


Figure 13.10 How filtering out centroids in the reducer step of MapReduce canopy clustering influences the dataset coverage by canopies. (left) When a centroid `c` is discarded because too close to one that has been chosen, part of the area covered by `c`'s canopy becomes uncovered for. (right) This can result in some points lying outside of any canopy.

The problem is that when we filter out any one of the centroids selected during the map step, a fraction of the points in its canopy might not be covered anymore, not even by the centroid that was chosen in its place: see the left side of figure 13.10, where the centroid marked as a star is drawn from the list of centroids, and as a result the other centroid within the inner radius T_2 from the selected one is filtered out - the area highlighted in red, however, is not covered by the canopy centered at the other centroid (the star).

While sometimes the lost coverage is made up for by other canopies, this isn't always the case: the right side of the same figure shows an example where a few points remain uncovered after a specific choice of centroids to keep.

There are several alternatives to solve this issue: